Reg. No. | | | | | | | | | | |

[12IS73]

SEVENTH SEMESTER B.E EXAMINATION DECEMBER–2015

DATA MINING

Time:3 Hours **Maximum Marks:70**

Instruction: Answer any **FIVE** full Questions.

1.	a.	What is data mining? Explain briefly data mining and knowledge discovery.	05
	b.	Explain in brief general characteristics of data sets.	05
	c.	List the processing steps to be applied to make the data more suitable for data mining.	04
2.	a.	Explain Tufte's principles for effective graphical display.	05
	b.	Explain the following terms:	09
		i. Stem and leaf plots	
		ii. Slicing and Dicing	
		iii. Roll-up and drill-down	
3.	a.	Explain Hunt's algorithm for inducing decision trees. And also mention its design issues.	07
	b.	Mention the characteristics of decision tree induction.	07
4.	a.	Consider a football game between two rival terms. Team o and team 1. Suppose team 0 wins 65% of time and team 1 wins the reaming matches. Among the games won by team 0, only 30% of them come from playing on team 1's football field. On the other hand, 75% of the victories for team 1 are obtained while playing at home. If team 1 are to host the next match between the two teams, which team will most likely emerge as winner?	07
	b.	Explain Bagging and Boosting algorithm.	07
5.		Explain apriori algorithm with an example.	14
6.	a.	Explain in detail different types of clustering.	07
	b.	Apply basic k-mean algorithm for the following data set: {7,10,11,75,2,111,42,50,48}. Assuming K=2.	03
	c.	Write a basic agglomerative hierarchical clustering algorithm and write Lance-Williams formula for cluster proximity.	04
7.	a.	Write DBSCAN algorithm.	05
	b.	List several important issues for cluster validation.	05
	c.	What is the relationship between cohesion and separation?	04
8.	a.	What are the applications of anomaly detection?	05
	b.	Along with a definition of density-based outlier detection. Write an algorithm for relative density outlier score algorithm and what are its strength and weaknesses.	09

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